Aminci Jonathan Adebanjo Gana

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github.com/HilbertSpaceExplorer

Summary

Mechanical Engineering B.Sc. student at Leibniz University Hannover with more than two years of hands-on experience in robotics, computer vision, sensor integration, and ML-based sensor data analysis. Seeking a working student position in machine learning, robotics, computer vision, or AI-based industrial systems.
Availability: 15–20 hours/week during the semester; full-time during semester breaks

Technical Skills

Computer Vision
OpenCV, stereo vision, RGB-D vision (Intel RealSense), ArUco/AprilTag detection, camera calibration, marker tracking, pose estimation

Sensors & Measurement
Sensor integration, force sensing, data acquisition, sensor calibration, experimental design, experimental validation

Modeling & Control
MATLAB/Simulink, data-driven modeling, model validation

Programming & Data Processing
Python (NumPy, Pandas, Matplotlib), Arduino C/C++, Java basics, JavaScript/TypeScript basics

Machine Learning
scikit-learn (classification, regression, model evaluation, data preprocessing), PyTorch basics, TensorFlow basics

Web Development
HTML, CSS, React/Next.js basics, Vercel deployment

Engineering & Prototyping
Autodesk Inventor, SolidWorks basics, technical drawing, 3D printing (Bambu Studio)

Tools
Git/GitHub, VS Code, Linux/Ubuntu, Microsoft Office

Languages

English
Native

German
C1

Mandarin
HSK 2 / A2

Spanish
A1/A2

Yoruba
B2 listening comprehension

Professional Experience

Student Assistant, Institute of Mechatronic Systems
04/2024 – Present
- Support research and student projects in automated driving, computer vision, and mechatronic systems.
- Develop and test Python/OpenCV-based perception workflows using ArUco/AprilTag markers, stereo vision, and Intel RealSense cameras.
- Assist with robotics laboratory experiments, including sensor integration, data acquisition, calibration, debugging, and evaluation.

Internship, Körber Feinmechanik GmbH
02/2023 – 04/2023
Gained hands-on experience in precision mechanics, manufacturing processes, and technical workshop operations.

Digital Marketing, Cerdana
05/2021 – 03/2022
Created and maintained digital content and performed structured analysis of relevant performance metrics.

Projects

Continuum Robot Modeling Under External Forces , Bachelor Thesis
- Modeling the shape of a continuum robot under external contact forces using tendon force sensors, base force/torque measurements, and stereo vision.
- Developed a vision-based method for determining the direction of an external force using an octagonal marker body, achieving an RMS reprojection error of ≤ 1.7 px.

Tools: MATLAB, Simulink, Python, ROS, Intel RealSense, OpenCV

ML-Based State Classification and Flow Estimation in a Compressed-Air System
- Developed and validated classification and regression models for industrial sensor data using Python and scikit-learn.
- Achieved 98.3% hold-out classification accuracy and a regression RMSE of 0.213 kW with $R^2 = 0.985$ using leave-one-experiment-out validation.
- Investigated feature selection, model complexity, and generalization performance to mitigate overfitting.

Tools: Python, scikit-learn, pandas, NumPy, classification, regression, sensor data analysis

Vision-Based Robustness Evaluation of PID Distance Control
- Developed a computer vision-based workflow to evaluate the robustness of a PID distance controller using recorded distance time-series data.
- Quantified control performance using the RMSE of the distance deviation from the 30 cm setpoint; qualitatively evaluated overshoot, oscillatory behavior, and settling response from the time series.

Tools: Python, OpenCV, NumPy, PID control, data analysis

Computer Vision-Based Lap Time Measurement for Line-Following Robots
- Developed a video-based timing system to automatically detect start/finish-line crossings.
- Automated robot lap-time measurement to reduce manual timing errors compared with hand-timed measurements.

Tools: Python, OpenCV, video processing

Education

B.Sc. Mechanical Engineering, Leibniz University Hannover
04/2023 – Expected 09/2026 | Hannover, Germany
- **Bachelor Thesis:** Identification of Dynamic Model Parameters of a Tendon-Driven Continuum Robot Under Contact Conditions
- **Focus Areas:** Mechatronics, Control Engineering, Measurement and Sensor Technology

Feststellungsprüfung, Niedersächsisches Studienkolleg
04/2022 – 02/2023 | Hannover, Germany
Final grade: 1.5

High School Diploma (WASSCE), Model Secondary School Maitama
11/2020 | Abuja, Nigeria
Qualification: WASSCE — 3 Distinctions & 6 Credits
German equivalent grade: 2.1

Volunteering & Leadership

Volunteering & Leadership
- Organized and led an online mathematics course, 2020
- Awareness programs, 2019